

Airport Management System

Behavioral Design Patterns Documentation

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Course: Software Modeling and Design

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Document Overview

This document presents the implementation and structural mapping of Behavioral Design Patterns within the Airport Management System (AMS). While previous models focused on dynamic state transitions and message sequencing, this document formally defines the static class architectures used to solve specific structural problems. It specifically details the Observer Pattern applied to bridge the Human Resources and System Administration modules.

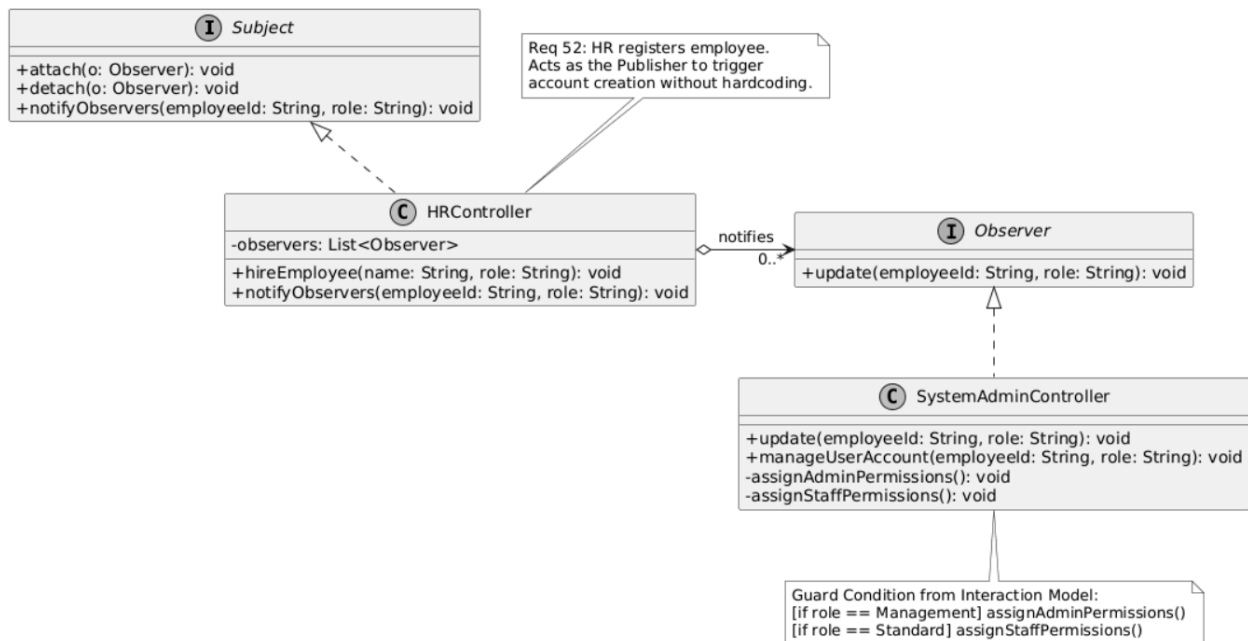
Objective

The primary goal of this document is to justify our architectural choices based on established software design principles. It provides strict UML Class Diagrams to demonstrate how the system fulfills its functional requirements while safely handling alternative flows and preventing tight coupling between independent subsystems.

Design Pattern 1: Observer Pattern (Module Integration: HR & System Admin)

- **Requirement Mapped:** Req 52 (HR registers employee records) and System Admin Account Creation.
- **Problem:** In our interaction models, when the Human Resources module (Ajla) registers a new employee, it needs to trigger the System Administration module (Eneida) to automatically generate a digital user account. If we hardcoded the System Admin classes directly into the HR code, the two modules would be tightly coupled, making the system fragile if administrative rules ever change.
- **Solution:** We applied the Behavioral **Observer Pattern**. The **HRController** acts as the Subject (or Publisher). When an employee is hired, it broadcasts a notification. The **SystemAdminController** acts as the Observer (or Subscriber), listening for that event to create the account.
- **Justification:** This pattern is ideal because it allows us to establish relations between objects at runtime. The HR backend does not need to know the internal implementation of the Admin backend. It also securely manages our conditional logic: when the Admin observer receives the notification, it uses our predefined guard conditions `[if role == Management]` or `[if role == Standard]` to assign the correct permissions.

2. The Class Diagram



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Document Overview

This document presents the implementation and structural mapping of Behavioral Design Patterns within the Airport Management System (AMS). While previous models focused on dynamic state transitions and message sequencing, this document formally defines the static class architectures used to solve specific structural problems. It specifically details the Facade Pattern used in the Flight Operations Module.

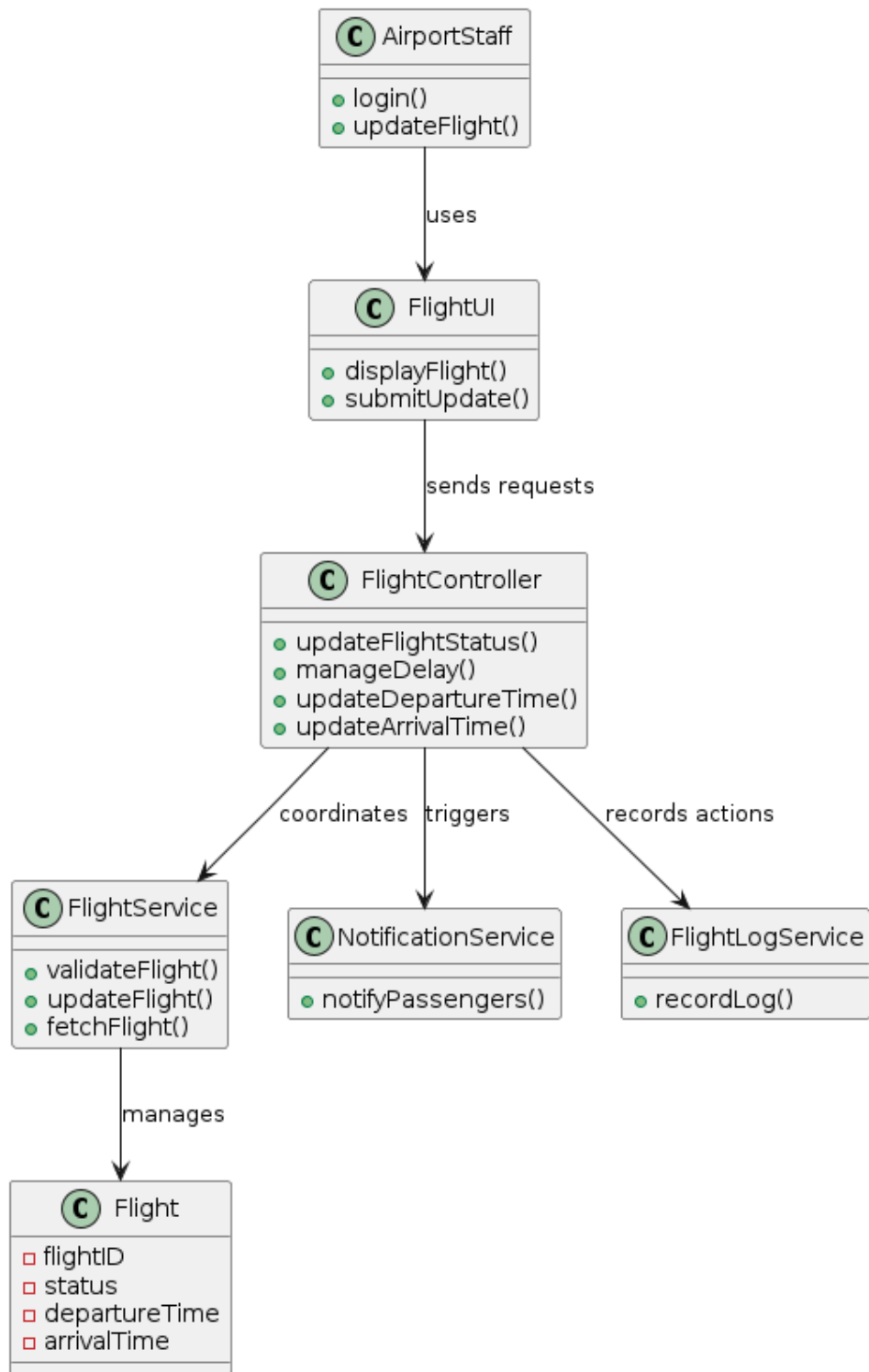
Objective

The primary goal of this document is to justify our architectural choices based on established software design principles. It provides strict UML Class Diagrams to demonstrate how the system fulfills its functional requirements while safely handling alternative flows and preventing tight coupling between independent subsystems.

Design Pattern 2: Facade Pattern (Module Integration: Airport Staff)

- **Requirements Mapped:**
 - Flight status management
 - Flight delay handling
 - Departure and arrival time updates
 - Passenger notification handling
 - Operational logging and database persistence
 - Centralized coordination of flight operations
- **Problem:** The Flight Operations Module contains multiple interconnected subsystems such as flight validation, scheduling, notifications, logging, and database management. If the user interface interacted directly with all these components, the system would become highly coupled, difficult to maintain, and more complex to extend. Any modification to one subsystem could require changes across several parts of the application.
- **Solution:** The system implements the **Facade Pattern** through the **FlightController** class. The controller acts as a centralized access point between the user interface and the internal operational subsystems. Instead of directly communicating with services such as **NotificationService** and **FlightLogService** the UI sends all operational requests to the **FlightController**, which coordinates the required actions internally.
- **Justification:** The Facade Pattern is appropriate for the Flight Operations Module because it simplifies communication with complex operational subsystems while reducing coupling between layers. The pattern improves maintainability by isolating subsystem changes from the user interface and centralizing operational workflows within the controller layer. It also enhances scalability, as additional operational services can be integrated behind the facade without requiring modifications to external components.

2. The Class Diagram



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Document Overview

This document presents the implementation and structural mapping of the Behavioral Chain of Responsibility Design Pattern within the Airport Management System (AMS). While previous system models focused on operational workflows and subsystem interactions, this document formally defines the request-handling architecture used to manage passenger service processes. It specifically details how passenger requests are sequentially processed through multiple handlers, including check-in, baggage registration, baggage tracking, and baggage claim services, without creating tight dependencies between system components.

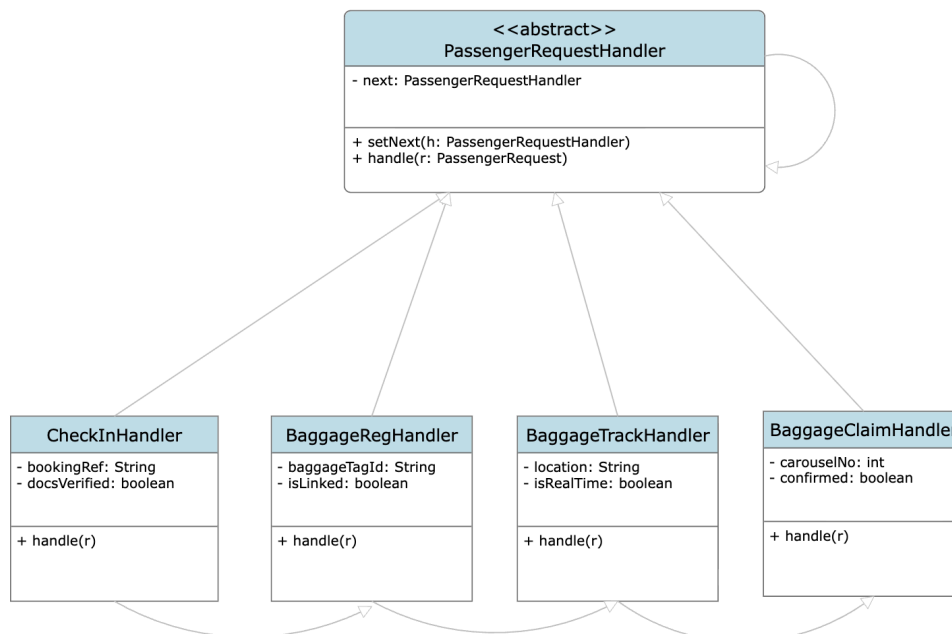
Objective

The primary goal of this document is to justify the architectural decisions behind the implementation of the Chain of Responsibility Pattern based on established software design principles. It provides UML Class Diagrams to demonstrate how the system efficiently processes passenger-related requests through independent handlers while ensuring flexibility, scalability, and maintainability. The approach prevents tight coupling between modules and supports future expansion of airport service functionalities.

Design Pattern 3: Chain of Responsibility Pattern (Passenger Request Handling)

- **Requirement Mapped:** Passenger service request processing within the Airport Management System, including check-in, baggage registration, baggage tracking, and baggage claim handling.
- **Problem:** In our Airport Management System, passenger requests can involve multiple services depending on the situation. For example, a passenger may first require check-in validation, then baggage registration, baggage tracking, or baggage claim confirmation. If all these responsibilities were hardcoded into one centralized controller, the system would become tightly coupled and difficult to maintain. Any modification in one process (e.g., baggage tracking rules) could negatively affect the entire request-handling workflow.
- **Solution:** We applied the Behavioral Chain of Responsibility Pattern. The abstract PassengerRequestHandler acts as the base handler and defines the request-processing chain through the setNext() and handle() methods. Each concrete handler (CheckInHandler, BaggageRegHandler, BaggageTrackHandler, and BaggageClaimHandler) is responsible for processing a specific passenger-related request. If a handler cannot fully process the request, it forwards it to the next handler in the sequence. This approach enables requests to move dynamically through the chain without hardcoding dependencies between modules.
- **Justification:** This pattern is ideal because it promotes loose coupling between request senders and handlers. Each class has a single responsibility, making the system easier to extend and maintain. For example, new passenger service handlers can be added without changing the existing logic. Additionally, the sequential handling process ensures that airport operations such as check-in, baggage registration, tracking, and claim management are processed in an organized and scalable manner.

2. The Class Diagram



Behavioral Design Patterns Documentation

Prepared by: Ani Velo

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Document Overview

This document presents the implementation and structural mapping of Behavioral Design Patterns within the **Security Management Module** of the Airport Management System (AMS). While previous models focused on dynamic message sequencing, this document formally defines the static class architectures used to solve specific security and validation problems. It specifically details the **Proxy Pattern** as they apply to passenger screening and emergency protocols.

Objective

The primary goal of this document is to justify our architectural choices based on established software design principles. It provides strict UML Class Diagrams to demonstrate how the security module fulfills its functional requirements while maintaining "thin" user interfaces and automated audit compliance.

Design Pattern 1: Proxy Pattern (Passenger Security Screening)

- **Requirement Mapped:** UC_SEC_01 (Passenger Security Screening).
- **Problem:** In the screening workflow, a **Security Officer** needs to access sensitive passenger data. Directly accessing the **PassengerProfile** entity without verification could lead to security breaches or data misuse.
- **Solution:** We applied the **Proxy Pattern**. The **SecurityController** acts as a protective Proxy. It intercepts requests from the **ScreeningUI** and only calls the **fetchProfile()** method on the real **PassengerProfile** after performing an internal **verifyStatus()** check.
- **Justification:** This pattern is ideal for security because it adds a layer of access control. It ensures that the "Entity-Layer Persistence" remains secure by only allowing authorized controllers to communicate with the database layer.

2. The Class Diagram

